Vibe Coding For AI Application Development

Teaching Record

24-07-2026

Scope

- Vibe Coding
  - What/Why/How
  - Tools
  - Python
  - IT Essentials
- AI Application Development
  - Workflow
  - Chatbot
  - Agents

Scope

- Overview of AI Applications with Vibe Coding
- Setup of tools
- Python with vibe coding
- Claude Code
- Use cases of AI applications
- Agentic use cases with vibe coding

Class One

- Overview of LLM Solutions
  - Prompts
  - Coding LLM
  - Agent
    - What is Agent
      - Execution
      - LLM (Brain)
      - Tools (Execution)
      - Memory
      - Harness (looping and orchestration)
- Setup of tools
  - Cursor
    - Subscription
  - Grok/Gemini (Browser mode LLM)
  - Claude code CLI (To be continued)
    - Cursor
    - Openrouter (LLM)
- Use sample py program to read it with coding agent

Class one

- Use case of using cursor
  - Read existing python script
  - Execute the python script
- Prompt techniques (To be continued)
- Python programming with vibe coding

Class two

- Recap of last lesson
  - **Grok (Browser)**
  - **Cursor**
    - **Write the program(s) in the working folder**
    - **Debug the program(s) in the working folder**
    - **Deploy the solution in cloud**
  - Claude code
- Prompt techniques
  - System prompt
  - User prompt
  - Zero shot
  - Few shot
- Python programming with vibe coding
  - Tools (Anaconda) (Jupyter notebook)
  - Data types
  - Function
  - **Data structures (To be continued)**
- **Class exercise**
  - **Vibe coding**
    - **Generate stock processing excel**